

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400539
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Dourval; Clement et al.

Aspirating smoke detector device operational analysis

Abstract

Methods, devices, and systems for an aspirating smoke detector device operational analysis are described herein. One device includes a memory, and a processor to execute executable instructions stored in the memory to receive operational information associated with an aspirating smoke detector device and generate, using the received operational information, an aspirating smoke detector device operational analysis including a real-time operational parameter associated with the aspirating smoke detector device, and a user interface configured to display the aspirating smoke detector device operational analysis.

Inventors: Dourval; Clement (Charlotte, NC), Sbardellini; Daniele (Padua, IT), Miheli; Mauro (Trieste, IT), Devaraja; Sandesh Hemmuru (Benguluru, IN), Gasparella; Simone (Trieste, IT), Piro; Domenico (Trieste, IT), Simeoni; Erika (Trieste, IT), Kaur; Reet (Ghaziabad, IN)

Applicant: Honeywell International Inc. (Charlotte, NC)

Family ID: 1000008779890

Assignee: Honeywell International Inc. (Charlotte, NC)

Appl. No.: 18/527518

Filed: December 04, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240194052 A1	Jun. 13, 2024

Related U.S. Application Data

continuation parent-doc US 17701925 20220323 US 11837074 child-doc US 18527518
continuation parent-doc US 17467965 20210907 US 11610472 20230321 child-doc US 17701925

Publication Classification

Int. Cl.: **G08B29/14** (20060101); **G06F3/0484** (20220101); **G08B17/10** (20060101); **H04W4/38** (20180101); **H04W4/80** (20180101)

U.S. Cl.:

CPC **G08B29/145** (20130101); **G06F3/0484** (20130101); **G08B17/10** (20130101); **H04W4/38** (20180201); **H04W4/80** (20180201);

Field of Classification Search

CPC: G08B (29/145); G06F (3/0484); H04W (4/38); H04W (4/80)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4173886	12/1978	Archbold et al.	N/A	N/A
10567445	12/2019	Logue et al.	N/A	N/A
10929735	12/2020	Zarakas et al.	N/A	N/A
10943444	12/2020	Boyd et al.	N/A	N/A
11143641	12/2020	Templeton	N/A	N/A
11231402	12/2021	Deininger	N/A	N/A
11610472	12/2022	Dourval	N/A	N/A
11761875	12/2022	Gasparella	356/338	G01N 15/06
11837074	12/2022	Dourval	N/A	G08B 29/145
2006/0220886	12/2005	Robertson	N/A	N/A
2006/0261967	12/2005	Marman	N/A	N/A
2009/0128327	12/2008	Lang	N/A	N/A
2010/0050744	12/2009	Petrovic	N/A	N/A
2010/0328082	12/2009	Danz et al.	N/A	N/A
2011/0074590	12/2010	Sacknoff et al.	N/A	N/A
2011/0161885	12/2010	Gonia et al.	N/A	N/A
2013/0213114	12/2012	Wetzig et al.	N/A	N/A
2014/0015680	12/2013	Chandler	N/A	N/A
2014/0318218	12/2013	Susel et al.	N/A	N/A
2016/0225251	12/2015	Di Marco et al.	N/A	N/A
2017/0154518	12/2016	Prabhakar et al.	N/A	N/A
2018/0301013	12/2017	Wang et al.	N/A	N/A
2019/0035253	12/2018	Jones, II et al.	N/A	N/A
2019/0124194	12/2018	Stolte	N/A	N/A
2019/0221101	12/2018	Golob	N/A	N/A
2019/0234924	12/2018	Anto et al.	N/A	N/A
2020/0226916	12/2019	Lim et al.	N/A	N/A
2020/0320857	12/2019	Lang et al.	N/A	N/A
2020/0363383	12/2019	Chow	N/A	N/A
2020/0402381	12/2019	Nelson	N/A	N/A
2021/0056834	12/2020	Wolff	N/A	N/A

2021/0104140	12/2020	Park et al.	N/A	N/A
2021/0142654	12/2020	Lobo	N/A	N/A
2021/0192913	12/2020	Via	N/A	N/A
2021/0259287	12/2020	Schmehl et al.	N/A	N/A
2021/0302267	12/2020	Tao et al.	N/A	N/A
2021/0318014	12/2020	Helbling	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201796460	12/2010	CN	N/A
102314745	12/2011	CN	N/A
204511522	12/2014	CN	N/A
208722373	12/2018	CN	N/A
2009238080	12/2008	JP	N/A

OTHER PUBLICATIONS

Wagner Brandschutzlosungen, “Wagner Diagnostic Tool DIAG3”, Nov. 1, 2014, pp. 1-15. cited by applicant

Wagner Brandschutzlosungen, “Installation Instructions Diagnostic Device for AD2-C11 Rev.d, Titanus 3000 and SuperSens”, Feb. 28, 1999, pp. 1-4. cited by applicant

Wagner Group GmbH; Air Sampling Smoke Detection System: Titanus Top Sens #174, Rev. A, Technical Manual, (Apr. 30, 2009) (214 pgs). cited by applicant

Primary Examiner: Tweel, Jr.; John A

Attorney, Agent or Firm: Brooks, Cameron & Huebsch, PLLC

Background/Summary

PRIORITY INFORMATION (1) This application is a Continuation of U.S. application Ser. No. 17/701,925, filed Mar. 23, 2022, which is a Continuation of U.S. application Ser. No. 17/467,965, filed Sep. 7, 2021, which issued as U.S. Pat. No. 11,610,472, on Mar. 21, 2023, the contents of which are incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to methods, devices, and systems for an aspirating smoke detector device operational analysis.

BACKGROUND

(2) Large facilities (e.g., buildings), such as commercial facilities, office buildings, hospitals, and the like, may have an alarm system that can be triggered during an emergency situation (e.g., a fire) to warn occupants to evacuate. For example, an alarm system may include a control panel (e.g., a fire control panel) and a plurality of aspirating smoke detector devices located throughout the facility (e.g., on different floors and/or in different rooms of the facility) that detect a hazard event, such as smoke generation (e.g., as the result of a fire or otherwise). The aspirating smoke detector can transmit a signal to the control panel in order to notify a building manager, occupants of the facility, emergency services, and/or others of the hazard event via alarms or other mechanisms.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an illustration of a display provided on a user interface showing an aspirating smoke detector device operational analysis, generated in accordance with one or more embodiments of the present disclosure.
- (2) FIG. 2 is an illustration of a display provided on a user interface showing a mobile device connecting with an aspirating smoke detector device, generated in accordance with one or more embodiments of the present disclosure.
- (3) FIG. 3 is an illustration of a display provided on a user interface showing an air flow rate and an air flow temperature of an aspirating smoke detector device, generated in accordance with one or more embodiments of the present disclosure.
- (4) FIG. 4 is an illustration of a display provided on a user interface showing a smoke level of an aspirating smoke detector device, generated in accordance with one or more embodiments of the present disclosure.
- (5) FIG. 5 is an illustration of a display provided on a user interface showing modification of a real-time operational parameter of an aspirating smoke detector device, generated in accordance with one or more embodiments of the present disclosure.
- (6) FIG. 6 is an illustration of a display provided on a user interface showing a log of events, generated in accordance with one or more embodiments of the present disclosure.
- (7) FIG. 7 is a mobile device and an aspirating smoke detector device for an aspirating smoke detector device operational analysis, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

- (8) Methods, devices, and systems for an aspirating smoke detector device operational analysis are described herein. One device includes a memory, and a processor to execute executable instructions stored in the memory to receive operational information associated with an aspirating smoke detector device and generate, using the received operational information, an aspirating smoke detector device operational analysis including a real-time operational parameter associated with the aspirating smoke detector device, and a user interface configured to display the aspirating smoke detector device operational analysis.
- (9) An aspirating smoke detector device can be utilized in a facility to detect a hazard event by detecting the presence of smoke. The aspirating smoke detector device can draw gas (e.g., air, via a blower) from the facility into a sensor through a network of pipes throughout the facility. The sensor can sample the gas in order to determine whether the gas includes smoke particles. In response to detection of smoke particles, the aspirating smoke detector device can transmit a signal to a control panel in the facility to signal detection of smoke particles.
- (10) An aspirating smoke detector device may monitor various operational parameters associated with the aspirating smoke detector device. For example, the aspirating smoke detector device may monitor a blower speed of a blower of the aspirating smoke detector device, an air flow rate of gas through the aspirating smoke detector device, an air flow temperature of gas through the aspirating smoke detector device, and/or a smoke level of gas through the aspirating smoke detector device, among other operational parameters associated with the aspirating smoke detector device.
- (11) Such operational parameters may provide insight to a user regarding the aspirating smoke detector device. For example, it may be beneficial for a user to monitor and/or review the operational parameters of the aspirating smoke detector device in order to determine a state of the aspirating smoke detector device, determine whether the aspirating smoke detector device may have detected smoke (e.g., related to a fire event or other event), predict issues relating to the aspirating smoke detector device and/or the aspirating smoke detection system in the facility, among other information.
- (12) Presenting such information via a user interface of a mobile device can allow a user to quickly

determine the status of an aspirating smoke detector device in the facility and generate awareness regarding facility safety. Further, the user may modify operational parameters of the aspirating smoke detector device via the user interface. Such presentation and modification functionality can allow for a robust but easy to understand presentation of hazard detection information.

(13) In the following detailed description, reference is made to the accompanying drawings that form a part hereof. The drawings show by way of illustration how one or more embodiments of the disclosure may be practiced.

(14) These embodiments are described in sufficient detail to enable those of ordinary skill in the art to practice one or more embodiments of this disclosure. It is to be understood that other embodiments may be utilized and that process, electrical, and/or structural changes may be made without departing from the scope of the present disclosure.

(15) As will be appreciated, elements shown in the various embodiments herein can be added, exchanged, combined, and/or eliminated so as to provide a number of additional embodiments of the present disclosure. The proportion and the relative scale of the elements provided in the figures are intended to illustrate the embodiments of the present disclosure, and should not be taken in a limiting sense.

(16) The figures herein follow a numbering convention in which the first digit or digits correspond to the drawing figure number and the remaining digits identify an element or component in the drawing. Similar elements or components between different figures may be identified by the use of similar digits. For example, **102** may reference element “**02**” in FIG. **1**, and a similar element may be referenced as **302** in FIG. **3**.

(17) FIG. **1** is an illustration of a display provided on a user interface **100** showing an aspirating smoke detector device operational analysis **102**, generated in accordance with one or more embodiments of the present disclosure. As illustrated in FIG. **1**, the aspirating smoke detector device operational analysis **102** can include real-time operational parameters **104-1**, **104-2** (referred to collectively herein as real-time operational parameters **104**). The real-time operational parameters **104** can include a blower speed **106**, an air flow rate **108**, and an air flow temperature **110**.

(18) The aspirating smoke detector device operational analysis **102** can be generated by a mobile device or a tablet (e.g., mobile device **726**, as described in connection with FIG. **7**) based on operational information associated with an aspirating smoke detector device. Operational information associated with the aspirating smoke detector device can be received by the mobile device from the aspirating smoke detector device located in a facility. Such operational information may be received via a network relationship between the aspirating smoke detector device and the mobile device, as is further described in connection with FIG. **2**.

(19) The operational information associated with the aspirating smoke detector device can be data relating to operation of the aspirating smoke detector device. For example, operational information can be data collected by various sensors included with the aspirating smoke detector device during operation of the aspirating smoke detector device. Such data may be utilized to illustrate various operational parameters of the aspirating smoke detector device, as is further described herein.

(20) The mobile device can generate, using the received operational information, the aspirating smoke detector device operational analysis **102**. As illustrated in FIG. **1**, the aspirating smoke detector device operational analysis **102** can include real-time operational parameters **104** associated with the aspirating smoke detector device, as is further described herein. As used herein, the term “real-time operational parameter” refers to a value relating to operation of a device that is generated within a specified timeframe so as to provide feedback regarding operation of the device. For example, the real-time operational parameters **104** may be values that describe operation of the aspirating smoke detector device, where such values are generated within a short timeframe so as to provide a sufficiently immediate indication of the operation of the aspirating smoke detector device.

(21) As illustrated in FIG. 1, the aspirating smoke detector device operational analysis **102** may include display of multiple real-time operational parameters **104**. The real-time operational parameter **104-1** can, for instance, be a blower speed **106** of a blower included in the aspirating smoke detector device. As mentioned above, the aspirating smoke detector device can utilize a blower to draw gas (e.g., air) from the facility into a sensor included in the aspirating smoke detector device. The blower can operate to draw gas into and cause gas to flow through the aspirating smoke detector device. The sensor included in the aspirating smoke detector device can sample the gas to detect the presence of smoke. Such gas can be drawn through a network of pipes throughout the facility into the sensor of the aspirating smoke detector device.

(22) The blower of the aspirating smoke detector device can operate at various speeds as indicated by the blower speed **106**. As indicated in FIG. 1, the blower of the aspirating smoke detector device may be operating at 2,800 rotations per minute (RPM).

(23) In some examples, the user interface **100** can display the real-time operational parameters **104** graphically on the user interface **100**. For example, the blower of the aspirating smoke detector device can be operating at 2,800 RPM and the blower speed can be indicated graphically on the user interface **100** as a half-circle bar graph. For instance, the lowest blower speed can be 0 RPM and the highest blower speed can be 7,000 RPM. The half-circle bar graph can be “filled” to around $\frac{1}{4}$ of the way full to indicate that the real-time blower speed is around $\frac{1}{4}$ (e.g., 2,800 RPM) of the highest blower speed (e.g., 7,000 RPM).

(24) In some examples, the user interface **100** can display the real-time operational parameters **104** as a numerical value on the user interface **100**. For example, the blower of the aspirating smoke detector device can be operating at 2,800 RPM and the blower speed can be indicated as a numerical value on the user interface **100** as 2,800 RPM.

(25) Although the aspirating smoke detector device operational analysis **102** is described above as including a blower speed **106**, embodiments of the present disclosure are not so limited. For example, the aspirating smoke detector device operational analysis **102** can include multiple real-time operational parameters **104**. For instance, the aspirating smoke detector device operational analysis **102** can further include an air flow rate **108** and/or an air flow temperature **110**.

(26) The air flow rate **108** can be a real-time operational parameter **104** included in the aspirating smoke detector device operational analysis **102** on the user interface **100**. As mentioned above, the blower of the aspirating smoke detector device can draw gas through the aspirating smoke detector device at different speeds, which can result in different air flow rates of the gas through the aspirating smoke detector device.

(27) In some examples, the user interface **100** can display the air flow rate graphically on the user interface **100**. For example, the air flow rate of the gas through the aspirating smoke detector device can be 45 liters per minute (L/min) and the air flow rate can be indicated graphically on the user interface **100** as a horizontal bar graph that can be “filled” in a direction to the left of the average flow rate to indicate the air flow rate through the aspirating smoke detector device is 45 L/min and is higher than the reference air flow rate of 40 L/min. In the center of the horizontal bar graph, the reference flow can be indicated via a vertical line on the horizontal bar graph. The aspirating smoke detector device can include a flow rate range, which can be variable. The flow rate range may be, in some examples, plus or minus 20%. For example, the reference flow in FIG. 1 is illustrated as being 40 L/min and the range can be minus 20% (e.g., 32 L/min) to plus 20% (e.g., 48 L/min). However, in other examples, the flow rate range may be plus or minus 50%. For example, the reference flow in FIG. 1 is illustrated as being 40 L/min and the range can be minus 50% (e.g., 20 L/min) to plus 50% (e.g., 60 L/min).

(28) In some examples, the user interface **100** can display the real-time operational parameters **104** as a numerical value on the user interface **100**. For example, the air flow rate of the gas through the aspirating smoke detector device can be 45 L/min and the air flow rate can be indicated as a numerical value on the user interface **100** as 45 L/min.

(29) The gas drawn through the aspirating smoke detector device can transit from a space in the facility to the detector and may be drawn at different air flow rates as a result of different blower speeds. Accordingly, the air flow temperature **110** of the gas can be a real-time operational parameter **104** included in the aspirating smoke detector device operational analysis **102** on the user interface **100**.

(30) Although not illustrated in FIG. **1** for clarity and so as not to obscure embodiments of the present disclosure, in some examples, the user interface **100** can display the air flow temperature graphically on the user interface **100**. For example, the air flow temperature of the gas through the aspirating smoke detector device can be 24 degrees Celsius ($^{\circ}$ C.) and the air flow temperature can be indicated graphically on the user interface **100** as a horizontal bar graph. For instance, the lowest air flow temperature can be -40° C. and the highest air flow rate can be 55° C. The horizontal bar graph can be “filled” to indicate the air flow temperature through the aspirating smoke detector device is 24° C.

(31) In some examples, the user interface **100** can display the real-time operational parameters **104** as a numerical value on the user interface **100**. For example, the air flow temperature of the gas through the aspirating smoke detector device can be 24° C. and the air flow temperature can be indicated as a numerical value on the user interface **100** as 24° C.

(32) As illustrated in FIG. **1**, the aspirating smoke detector device operational analysis **102** can include a plurality of real-time operational parameters **104** associated with the aspirating smoke detector device. The aspirating smoke detector device operational analysis **102** can further include, in some examples, threshold levels **116-1**, **116-2** (referred to collectively herein as threshold levels **116**) associated with each real-time operational parameter **104**.

(33) The air flow rate **108** can include threshold levels **116-1**, **116-2** shown on the horizontal bar graph. The horizontal bar graph can include a first threshold level **116-1** and a second threshold level **116-2**. The first threshold level **116-1** can be, for instance, a lower threshold level of 20 L/min and the second threshold level **116-2** can be a higher threshold level of 60 L/min. In an example in which the real-time air flow rate **108** of the aspirating smoke detector device exceeds the first threshold level **116-1** (e.g., the lower threshold level) or the second threshold level **116-2** (e.g., the higher threshold level), such instance would be displayed graphically on the user interface **100**. The real-time operational parameters **104** exceeding a threshold level **116** may indicate an error or problem with the aspirating smoke detector device, the piping network of the aspirating smoke detector device, or any other event which may be useful for a user of the mobile device displaying the user interface **100** to know about. Such a scenario may be displayed on the user interface **100**, as is further described in connection with FIGS. **3** and **4**.

(34) FIG. **2** is an illustration of a display provided on a user interface **200** showing a mobile device connecting with an aspirating smoke detector device **214-1**, **214-2**, **214-N**, generated in accordance with one or more embodiments of the present disclosure. The user interface can display various aspirating smoke detector devices **214-1**, **214-2**, **214-N** (referred to collectively herein as aspirating smoke detector devices **214**).

(35) As illustrated in FIG. **2**, the mobile device can connect to various different aspirating smoke detector devices **214**. The mobile device may display, on the user interface **200**, aspirating smoke detector devices **214** which are within a threshold distance of the mobile device. As illustrated in FIG. **2**, a user may select aspirating smoke detector device **214-1** to connect to (e.g., as illustrated in FIG. **2** as a “cursor” in the shape of a hand with an outstretched finger).

(36) The mobile device may connect to the aspirating smoke detector device **214-1** via a network relationship. The network relationship may be a wired and/or wireless network relationship. Examples of such a network relationship can include a local area network (LAN), wide area network (WAN), personal area network (PAN), a distributed computing environment (e.g., a cloud computing environment), storage area network (SAN), Metropolitan area network (MAN), a cellular communications network, Long Term Evolution (LTE), visible light communication

(VLC), Bluetooth, Worldwide Interoperability for Microwave Access (WiMAX), Near Field Communication (NFC), infrared (IR) communication, Public Switched Telephone Network (PSTN), radio waves, and/or the Internet, among other types of network relationships.

(37) As mentioned above, the mobile device may connect to the aspirating smoke detector device via a wireless connection. In some examples, the wireless connection can be a Bluetooth connection.

(38) As illustrated in FIG. 2, upon selection of the aspirating smoke detector device **214-1**, the user interface **200** can display various attributes about the aspirating smoke detector device **214-1**. Such attributes may include an identification number for the aspirating smoke detector device **214-1**, its operational mode, its location, its pipe configuration, its pipe length, and/or when the mobile device was last connected with the aspirating smoke detector device **214-1**. The mobile device may connect to the aspirating smoke detector device in response to a user selecting the “Pair Device” button via the user interface **200**. However, embodiments of the present disclosure are not so limited. For instance, in some examples, the mobile device may automatically pair with the aspirating smoke detector device **214-1** when the mobile device is within a threshold distance of the aspirating smoke detector device **214-1**.

(39) In response to the mobile device being connected to the aspirating smoke detector device, the mobile device can receive the operational information. For example, when the mobile device connects with the aspirating smoke detector device **214-1**, the aspirating smoke detector device **214-1** can transmit the operational information to the mobile device. In response, the mobile device can generate the aspirating smoke detector device operational analysis for the aspirating smoke detector device **214-1**.

(40) FIG. 3 is an illustration of a display provided on a user interface **300** showing an air flow rate **308** and an air flow temperature **310** of an aspirating smoke detector device, generated in accordance with one or more embodiments of the present disclosure. As illustrated in FIG. 3, at a first time, the aspirating smoke detector device operational analysis **302-1** may change at a second time to be aspirating smoke detector device operational analysis **302-2**.

(41) As previously described in connection with FIGS. 1 and 2, the mobile device can receive operational information associated with an aspirating smoke detector device in response to a wireless connection being established with the aspirating smoke detector device and generate, using the received operational information, the aspirating smoke detector device operational analysis **302**. The aspirating smoke detector device operational analysis **302-1**, **302-2** can include real-time operational parameters associated with the aspirating smoke detector device at different times, including an air flow rate **308** of gas through the aspirating smoke detector device and an air flow temperature **310** of the gas through the aspirating smoke detector device.

(42) The aspirating smoke detector device may include two flow channels for gas through the aspirating smoke detector device. The two channels can be displayed on the user interface **300** as “Channel 1” and “Channel 2”. During the aspirating smoke detector device operational analysis **302-1**, Channel 1 may include an associated air flow rate **308** of 45 L/min, and Channel 2 may include an associated air flow rate **308** of 25 L/min.

(43) As illustrated in FIG. 3, the air flow rate **308** through Channel 1 of the aspirating smoke detector device can be 45 L/min, and can be displayed graphically (e.g., via the horizontal bar graph) and as a numerical value of 45 L/min. Additionally, the air flow rate **308** through Channel 2 of the aspirating smoke detector device can be 25 L/min, and can be displayed graphically (e.g., via the horizontal bar graph) and as a numerical value of 25 L/min.

(44) Additionally, Channel 1 may include an associated air flow temperature of 24° C., and Channel 2 may include an air flow temperature of 24° C. The air flow temperatures of Channel 1 and Channel 2 can be displayed as numerical values of 24° C.

(45) As indicated in the aspirating smoke detector device operational analysis **302-1**, the air flow rate **308** in both Channels 1 and 2 may include a first threshold level **316-1** and a second threshold

level **316-2**. As illustrated in FIG. 3, the air flow rates **308** through both Channel 1 and Channel 2 are within the threshold range defined by the first (e.g., lower) threshold level **316-1** and the second (e.g., upper) threshold level **316-2**.

(46) In some examples, the mobile device can color code the real-time operational parameters in the aspirating smoke detector device operational analyses **302**. For example, the air flow rate in Channels 1 and 2 may be color coded as green in the horizontal bar graph to indicate the air flow rates do not exceed the first threshold level **316-1** or the second threshold level **316-2** and are within a threshold range.

(47) After a period of time has elapsed since the aspirating smoke detector device operational analysis **302-1** was generated, the real-time operational parameters may change. For example, during the aspirating smoke detector device operational analysis **302-2**, Channel 1 may include an associated air flow rate **308** of 75 L/min, whereas Channel 2 may include an associated air flow rate **308** of 25 L/min.

(48) As illustrated in the aspirating smoke detector device operational analysis **302-2**, the air flow rate **308** through Channel 1 exceeds the second threshold level **316-2**. As a result, the air flow rate **308** through Channel 1 may no longer be within the threshold range defined by the first threshold level **316-1** and the second threshold level **316-2**.

(49) The mobile device can color code the air flow rate **308** through Channel 1 in the aspirating smoke detector device operational analyses **302-2** differently based on the air flow rate **308** exceeding the second threshold level **316-2**. For example, the air flow rate in Channel 1 may be color coded as red in the horizontal bar graph to indicate the air flow rate **308** through Channel 1 exceeds the second threshold level **316-2** and is outside the threshold range. The air flow rate in Channel 2 may still be color coded as green in the horizontal bar graph to indicate the air flow rate **308** through Channel 2 does not exceed the first threshold level **316-1** or the second threshold level **316-2**.

(50) Based on a real-time operational parameter (e.g., the air flow rate **308** in Channel 1) exceeding a threshold level (e.g., threshold level **316-2**), the mobile device can generate an alert. The alert may be displayed on the user interface **300**. In some examples, the alert may be an audible alert emitted by a speaker of the mobile device, the alert may be a tactile alert caused by vibration of the mobile device, etc. In some examples, the alert may be transmitted to another computing device to alert the user and/or other users (e.g., in the form of a text message, email, etc.).

(51) FIG. 4 is an illustration of a display provided on a user interface **400** showing a smoke level **412** of an aspirating smoke detector device, generated in accordance with one or more embodiments of the present disclosure. As illustrated in FIG. 4, at a first time, the aspirating smoke detector device operational analysis **402-1** may change at a second time to be aspirating smoke detector device operational analysis **402-2**.

(52) Similar to FIG. 3, the mobile device can receive operational information associated with an aspirating smoke detector device and generate the aspirating smoke detector device operational analyses **402-1**, **402-2**. The aspirating smoke detector device operational analysis **402-1**, **402-2** can include real-time operational parameters associated with the aspirating smoke detector device at different times, including a smoke level **412**.

(53) The smoke level **412** can be a real-time operational parameter included in the aspirating smoke detector device operational analysis **402-1**, **402-2** on the user interface **400**. As previously described herein, the blower of the aspirating smoke detector device can draw gas through the aspirating smoke detector device, which can be sampled for smoke particles.

(54) In some examples, the user interface **400** can display the smoke level **412** graphically on the user interface **400**. For example, the smoke level of the gas through the aspirating smoke detector device can be smoke level 1 (e.g., in Channel 1) and smoke level 3 (e.g., in Channel 2) and the smoke levels can be indicated graphically on the user interface **400** as a horizontal bar graph. For instance, the lowest smoke level can be 0 and the highest smoke level can be 9. The horizontal bar

graph can be “filled” in a direction to the left of the 0 smoke level to indicate the smoke level of the gas through the aspirating smoke detector device as being smoke level 1 in Channel 1 and smoke level 3 in Channel 2.

(55) In some examples, the user interface **400** can display the real-time operational parameters as a numerical value on the user interface **400**. For example, the smoke level of the gas through Channel 1 can be smoke level 1 and the smoke level can be indicated as a numerical value on the user interface **400** as Smoke Level 1. Similarly, the smoke level of the gas through Channel 2 can be indicated as a numerical value on the user interface **400** as Smoke Level 03.

(56) As illustrated in FIG. 4, the smoke levels **412** can include a first threshold level **418-1** for an action alert and a second threshold level **418-2** for a fire alert associated with the smoke levels **412**. Different thresholds can be set based on an environment the aspirating smoke detector device is installed. For instance, in a dirty environment with a lot of dust, the threshold levels **418** can be set to a lower sensitivity, whereas in a cleaner environment relative to the dirty environment, the threshold levels **418** can be set to a higher sensitivity relative to the dirty environment. The first threshold level **418-1** can be smoke level 2 and the second threshold level **418-2** can be smoke level 4. In the aspirating smoke detector device operational analysis **402-1**, the smoke level **412** in Channel 1 can be smoke level 1, which does not exceed the first threshold level **418-1**. The mobile device can, in some examples, color code the horizontal bar graph green based on the smoke level in Channel 1 not exceeding the first threshold level **418-1**. Color coding the smoke level in Channel 1 green can indicate to a user the smoke level through Channel 1 is not of concern, as there is little to no smoke present in the space in the facility.

(57) Additionally in the aspirating smoke detector device operational analysis **402-1**, the smoke level in Channel 2 can be smoke level 3, which does exceed the first threshold level **418-1** but not the second threshold level **418-2**. The mobile device can, in some examples, color code the horizontal bar graph orange based on the smoke level in Channel 2 exceeding the first threshold level **418-1** but not the second threshold level **418-2**. Color coding the smoke level in Channel 2 orange can indicate to a user the smoke level through Channel 2 is more of a concern, as there may be some amount of smoke present in the space in the facility. In some examples, the mobile device may not generate an alert in response to the smoke level exceeding the first threshold level **418-1** but not the second threshold level **418-2**. In response to the smoke level exceeding the first threshold level **418-1**, the aspirating smoke detector device is to generate an action alert, and in response to the smoke level exceeding the second threshold level **418-2**, the aspirating smoke detector device is to generate a fire alarm. In some examples, the mobile device may also generate an alert in response to the smoke level exceeding the first threshold level **418-1** and/or the second threshold level **418-2**.

(58) In the aspirating smoke detector device operational analysis **402-2**, the smoke level **412** in Channel 1 can be smoke level 1, which does not exceed the first threshold level **418-1**. The mobile device can, in some examples, color code the horizontal bar graph green based on the smoke level in Channel 1 not exceeding the first threshold level **418-1**. Color coding the smoke level in Channel 1 green can indicate to a user the smoke level through Channel 1 is not of concern, as there is little to no smoke present in the space in the facility.

(59) Additionally in the aspirating smoke detector device operational analysis **402-2**, the smoke level in Channel 2 can be smoke level 7, which exceeds both the first threshold level **418-1** and the second threshold level **418-2**. The mobile device can, in some examples, color code the horizontal bar graph red based on the smoke level in Channel 2 exceeding the first threshold level **418-1** and the second threshold level **418-2**. Color coding the smoke level in Channel 2 red can indicate to a user the smoke level through Channel 2 is of concern, as there may be a fire related event occurring that is generating smoke in the space in the facility. In some examples, the mobile device may generate an alert in response to the smoke level exceeding the first threshold level **418-1** and the second threshold level **418-2**, and the alert may be the same type of alert or a different type of alert

as the alert generated in response to the smoke level exceeding the first threshold level **418-1** but not the second threshold level **418-2**.

(60) FIG. 5 is an illustration of a display provided on a user interface **500** showing modification of a real-time operational parameter of an aspirating smoke detector device, generated in accordance with one or more embodiments of the present disclosure. As illustrated in FIG. 5, the real-time operational parameter can be a blower speed **506**.

(61) In some examples, a user may utilize the mobile device to modify a real-time operational parameter. For example, a user may wish to modify a blower speed of a blower of the aspirating smoke detector device.

(62) As illustrated in FIG. 5, the blower speed may be speed level 6. Such a speed level of 6 may correspond to a blower speed of 2,800 RPM (e.g., as previously described in connection with FIG. 1). Accordingly, the user may wish to increase the blower speed to 3,000 RPM. Accordingly, the user may increase the blower speed level to 8.

(63) The user may increase the blower speed level to 8 via the user interface **500** of the mobile device. For example, the mobile device can receive an input to modify the real-time operational parameter (e.g., the blower speed). The mobile device may receive the input via a touch-screen display. The mobile device can cause the aspirating smoke detector device to modify the real-time operational parameter in response to the user interface receiving the input. For example, the mobile device can receive the input at the modify parameter **520** button displayed on the user interface **500** to change the blower speed level from 6 to 8, and the mobile device can cause the aspirating smoke detector device to increase the blower speed from 2,800 RPM to 3,000 RPM.

(64) Although the blower speed is described above as being modified by changing a blower speed level, embodiments of the present disclosure are not so limited. For example, the user may directly change, via a user input, the blower speed from 2,800 RPM to 3,000 RPM via a different modify parameter **520** button than is illustrated in FIG. 5. Such a different modify parameter **520** button may be a “slider”, a numerical input (e.g., a user types in 3,000 via a keyboard/number pad input displayed on the user interface **500**), etc.

(65) Additionally, although the blower speed is described above as the real-time operational parameter being modified, embodiments of the present disclosure are not so limited. For example, a user may change other real-time operational parameters (e.g., air flow rate, device sensitivity, etc.).

(66) FIG. 6 is an illustration of a display provided on a user interface showing a log **624** of events **622**, generated in accordance with one or more embodiments of the present disclosure. The log **624** may be generated by a mobile device.

(67) As illustrated in FIG. 6, the mobile device can generate a log of the received operational information over time. The received operational information can indicate to a user operational status of the aspirating smoke detector device in the form of events **622-1**, **622-2**, **622-3**, **622-N** (referred to collectively herein as events **622**). For example, event **622-1** can be a low air flow alert, indicating the air flow rate of the gas through a channel of the aspirating smoke detector device may be low. A user may then investigate the cause of the low air flow rate. Other events **622** may include an event **622-2** indicating a battery powering the aspirating smoke detector device has been discharged, an event **622-3** indicating an air flow fault, an event **622-N** indicating the voltage of the aspirating smoke detector device is low, etc. Such events **622** may be determined from the operational information by the mobile device and displayed on the user interface **600** via the log **624**.

(68) In some examples, the mobile device may generate a report associated with the log **624**. For example, the report may include a description of the operation of the aspirating smoke detector device over a time period, including the log **624** of the events **622**, and may be exportable for printing, presentation, etc. In some examples, the report may be displayed on the user interface **600**.

(69) An aspirating smoke detector device operational analysis, according to the present disclosure, can allow a user to quickly determine a status of an aspirating smoke detector device, reducing a

user's cognitive workload and increasing efficiency. Further, an aspirating smoke detector device operational analysis can allow for problems arising in the aspirating smoke detector device and/or associated network of piping to be identified and fixed, ensuring proper operation of the aspirating smoke detector device in the case of a fire event in the facility.

(70) FIG. 7 is a mobile device **726** and an aspirating smoke detector device **714** for an aspirating smoke detector device operational analysis, in accordance with one or more embodiments of the present disclosure. As illustrated in FIG. 7, the mobile device **726** can include a user interface **700**, memory **730**, and a processor **728** for an aspirating smoke detector device operational analysis in accordance with the present disclosure.

(71) The mobile device **726** can be, for example, a device that is (or can be) carried and/or worn by a user. For example, the mobile device **726** can be a phone (e.g., a smart phone), a tablet, a personal digital assistant (PDA), smart glasses, and/or a wrist-worn device (e.g., a smart watch), among other types of mobile devices.

(72) The memory **730** can be any type of storage medium that can be accessed by the processor **728** to perform various examples of the present disclosure. For example, the memory **730** can be a non-transitory computer readable medium having computer readable instructions (e.g., computer program instructions) stored thereon that are executable by the processor **728** for an aspirating smoke detector device operational analysis in accordance with the present disclosure. The computer readable instructions can be executable by the processor **728** to redundantly generate the aspirating smoke detector device operational analysis.

(73) The memory **730** can be volatile or nonvolatile memory. The memory **730** can also be removable (e.g., portable) memory, or non-removable (e.g., internal) memory. For example, the memory **730** can be random access memory (RAM) (e.g., dynamic random access memory (DRAM) and/or phase change random access memory (PCRAM)), read-only memory (ROM) (e.g., electrically erasable programmable read-only memory (EEPROM) and/or compact-disc read-only memory (CD-ROM)), flash memory, a laser disc, a digital versatile disc (DVD) or other optical storage, and/or a magnetic medium such as magnetic cassettes, tapes, or disks, among other types of memory.

(74) Further, although memory **730** is illustrated as being located within mobile device **726**, embodiments of the present disclosure are not so limited. For example, memory **730** can also be located internal to another computing resource (e.g., enabling computer readable instructions to be downloaded over the Internet or another wired or wireless connection).

(75) The mobile device **726** can be connected to an aspirating smoke detector device **714**. As previously described in connection with FIG. 2, the mobile device **726** can be wirelessly connected to the aspirating smoke detector device **714**. In some examples, the mobile device **726** can be wirelessly connected to the aspirating smoke detector device **714** via a Bluetooth connection.

(76) As illustrated in FIG. 7, the mobile device **726** includes a user interface **700**. For example, the user interface **700** can display the aspirating smoke detector device operational analysis (e.g., as previously described in connection with FIGS. 1-6) in a display. A user (e.g., operator) of the mobile device **726** can interact with the mobile device **726** via the user interface **700**. For example, the user interface **700** can provide (e.g., display and/or present) information to the user of mobile device **726**, and/or receive information from (e.g., input by) the user of mobile device **726**. For instance, in some embodiments, user interface **700** can be a graphical user interface (GUI) that can provide and/or receive information to and/or from the user of the mobile device **726**. The display can be, for instance, a touch-screen (e.g., the GUI can include touch-screen capabilities).

Alternatively, a display can include a television, computer monitor, mobile device screen, other type of display device, or any combination thereof, connected to the mobile device **726** and configured to receive a video signal output from the mobile device **726**.

(77) The user interface **700** can be localized to any language. For example, the user interface **700** can display the aspirating smoke detector device operational analysis in any language, such as

English, Spanish, German, French, Mandarin, Arabic, Japanese, Hindi, etc.

(78) Although specific embodiments have been illustrated and described herein, those of ordinary skill in the art will appreciate that any arrangement calculated to achieve the same techniques can be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments of the disclosure.

(79) It is to be understood that the above description has been made in an illustrative fashion, and not a restrictive one. Combination of the above embodiments, and other embodiments not specifically described herein will be apparent to those of skill in the art upon reviewing the above description.

(80) The scope of the various embodiments of the disclosure includes any other applications in which the above structures and methods are used. Therefore, the scope of various embodiments of the disclosure should be determined with reference to the appended claims, along with the full range of equivalents to which such claims are entitled.

(81) In the foregoing Detailed Description, various features are grouped together in example embodiments illustrated in the figures for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the embodiments of the disclosure require more features than are expressly recited in each claim.

(82) Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

Claims

1. A mobile device for an aspirating smoke detector device operational analysis, comprising: a memory; and a processor configured to execute executable instructions stored in the memory to: receive operational information associated with an aspirating smoke detector device in response to the mobile device wirelessly connecting to the aspirating smoke detector device; and generate, using the received operational information, an operational analysis of the aspirating smoke detector device, wherein the operational analysis indicates an operational status of the aspirating smoke detector device; and a user interface configured to display the aspirating smoke detector device operational analysis.
2. The mobile device of claim 1, wherein the processor is configured to execute the instructions to send a command to the aspirating smoke detector device.
3. The mobile device of claim 1, wherein the operational analysis includes a blower speed of the aspirating smoke detector device.
4. The mobile device of claim 3, wherein the blower speed is displayed as at least one of: graphically on the user interface; and a numerical value on the user interface.
5. The mobile device of claim 1, wherein the operational analysis includes an air flow rate of gas through the aspirating smoke detector device.
6. The mobile device of claim 5, wherein the air flow rate is displayed as at least one of: a numerical value on the user interface; graphically on the user interface; and color coded on the user interface based on a value of the air flow rate.
7. The mobile device of claim 1, wherein the operational analysis includes a smoke level of gas through the aspirating smoke detector device.
8. The mobile device of claim 7, wherein the smoke level is displayed as at least one of: graphically on the user interface; a numerical value on the user interface; and color coded on the user interface based on a value of the smoke level.
9. The mobile device of claim 1, wherein the user interface is configured to display a number of aspirating smoke detector devices, wherein each of the number of aspirating smoke detector devices are within a threshold distance of the mobile device.

10. A non-transitory computer readable medium having computer readable instructions stored thereon that are executable by a processor to: display, on a user interface of a mobile device, a number of aspirating smoke detector devices, wherein each of the number of aspirating smoke detector devices are within a threshold distance of the mobile device; receive, via the user interface, a selection of one of the number of aspirating smoke detector devices; wirelessly connect to the selected aspirating smoke detector device; and send a command from the mobile device to the selected aspirating smoke detector device.
 11. The non-transitory computer readable medium of claim 10, wherein the command is a command to set an alarm threshold of the aspirating smoke detector device.
 12. The non-transitory computer readable medium of claim 10, wherein the command is a command to set a sensitivity of the aspirating smoke detector device.
 13. The non-transitory computer readable medium of claim 10, wherein the command is a command to set a blower speed of the aspirating smoke detector device.
 14. The non-transitory computer readable medium of claim 10, wherein the computer readable instructions are executable by the processor to: generate an alert in response to an event experienced by the aspirating smoke detector device; and display the alert on the user interface.
 15. A mobile device for an aspirating smoke detector device operational analysis, comprising: a memory; a processor configured to execute executable instructions stored in the memory to: receive operational information associated with an aspirating smoke detector device in response to the mobile device wirelessly connecting to the aspirating smoke detector device; generate, using the received operational information, an operational analysis of the aspirating smoke detector device including a log of an event experienced by the aspirating smoke detector device; and a user interface configured to display the operational analysis.
 16. The mobile device of claim 15, wherein the event experienced by the aspirating smoke detector device is a fault associated with the aspirating smoke detector device.
 17. The mobile device of claim 16, wherein the fault associated with the aspirating smoke detector device is a fault associated with an air flow rate of the aspirating smoke detector device.
 18. The mobile device of claim 15, wherein the operational analysis indicates an operational status of the aspirating smoke detector device.
 19. The mobile device of claim 15, wherein the mobile device is configured to wirelessly connect to the aspirating smoke detector device via Bluetooth.
 20. The mobile device of claim 15, wherein the mobile device is a smart phone.
-